A Helmholtz Resonator Model for Gas-Pocket Mechanotransduction in the Human Abdomen: Theoretical Framework and Sensitivity Analysis

Jonathan Mace[1], Brian R. Mace[2]

[1]Microsoft Research, Redmond, WA, USA

[2]Department of Mechanical and Mechatronics Engineering, University of Auckland, New Zealand

Correspondence: jonathanmace@microsoft.com

March 29, 2026

Abstract

Whole-body acoustic models predict negligible tissue displacement from airborne infrasound, yet gastrointestinal (GI) discomfort is anecdotally reported at high sound pressure levels (SPL). We develop a first-principles model to investigate whether *intraluminal gas pockets* could act as impedance-matching transducers, converting pressure oscillations to local wall strain more efficiently than whole-cavity resonance. A constrained-bubble model—Minnaert dynamics modified for the elastic intestinal wall and cylindrical lumen geometry—predicts that gas pockets of 5 mL to 100 mL have resonant frequencies from tens to hundreds of hertz, depending on geometry, but sub-resonantly amplify tissue displacement as compliant inclusions in an otherwise stiff medium. Under idealised sealed-pocket conditions, the model predicts SPL thresholds for illustrative mechanotransduction activation ranging from 111 dB (100 mL pocket) to 120 dB (5 mL). However, the analysis identifies acoustic short-circuiting via the GI air column as a critical limitation: at 7 Hz, pressure equalisation attenuates the effective driving pressure by up to 83%, potentially reducing all displacements below activation thresholds. Whether this short circuit is suppressed *in vivo*—by mucosal folds, liquid plugs, or luminal collapse—remains the key unknown. A sensitivity analysis ($N = 10\,000$ Monte Carlo samples) identifies gas pocket volume and acoustic sealing as the dominant parameters. We define the experimental measurements needed to test this hypothesis, including *in vivo* pocket impedance characterisation and tissue-level mechanotransduction thresholds.

Keywords: infrasound, bowel gas, constrained bubble, mechanotransduction, Piezo channels, gastrointestinal acoustics, Minnaert resonance, hypothesis, acoustic short circuit

1 Introduction

The so-called "brown note"—a hypothetical infrasonic frequency that induces involuntary bowel movements—has persisted in popular culture and military speculation for decades, despite a lack of

credible experimental evidence [Jauchem and Cook, 2007, Leventhall, 2007, Parker and Stone, 2000, Savage and Hyneman, 2005]. Early aerospace-medicine studies documented nausea, abdominal vibration, and vague GI discomfort in subjects exposed to high-intensity infrasound (>140 dB), but attributed these effects primarily to whole-body vibration rather than airborne acoustic coupling [Mohr et al., 1965, von Gierke, 1974].

Our companion study [Mace et al., 2026] showed that airborne infrasound cannot excite abdominal-cavity flexural resonances at physiologically meaningful amplitudes. The breathing mode of the abdomen—often conflated with the brown-note myth in popular accounts—lies at ~2490 Hz, far above the infrasound range. The relevant flexural modes ($n \geq 2$) do fall at 4 Hz to 10 Hz, but the $(ka)^n$ coupling penalty renders airborne excitation negligible: the required SPL for 1 µm tissue displacement exceeds 180 dB. The present paper investigates the one mechanism that the whole-cavity model omits: the small pockets of gas that the gastrointestinal tract harbours with neither apology nor predictability.

However, the whole-cavity analysis treats the abdomen as a homogeneous elastic shell filled with incompressible fluid. This neglects a critical feature: the human gastrointestinal tract normally contains 100 mL to 200 mL of gas distributed in multiple discrete pockets [Levitt, 1971, Suarez et al., 1997], with pathological or post-prandial values reaching 500 mL to 1500 mL [Bedell et al., 1956, Serra et al., 2001]. These gas pockets constitute *compressible inclusions* within an otherwise nearly incompressible medium, creating local impedance mismatches of ~ 3600:1 at each gas–tissue interface.

We hypothesise that bowel gas pockets could function as *acoustic transducers*: the spatially uniform infrasonic pressure field would drive radial oscillation of each gas–tissue interface, converting acoustic pressure to local wall strain. This mechanism would be fundamentally different from—and potentially far more efficient than—whole-cavity resonance, because:

1. Gas compressibility provides a compliance that tissue lacks;
2. The transduction is broadband (not limited to a single resonance);
3. The displacement is concentrated at the gas–tissue boundary, where mechanosensitive ion channels (Piezo1, Piezo2) are expressed in enterochromaffin cells and enteric neurons [Coste et al., 2010, Alcaino et al., 2017, Zeitzschel and Lechner, 2024].

In this paper we develop a constrained-bubble model for intraluminal gas pockets, estimate the conditions under which Piezo channel activation thresholds could potentially be reached, identify the acoustic short-circuit as the critical limiting factor, and use sensitivity analysis to determine which parameters most affect the outcome. The aim is not to demonstrate that this mechanism operates *in vivo*, but to establish a theoretical framework, identify the key unknowns, and define the experimental measurements that would resolve the question.

2 Model

2.1 Geometry and assumptions

We model a single gas pocket as a compressible inclusion within the intestinal lumen, surrounded by an elastic wall of thickness h_w, Young's modulus E_w, Poisson's ratio ν_w, and density ρ_w

(Fig. 1). The surrounding tissue is modelled as an incompressible fluid of density ρ_f at infrasound wavelengths ($\lambda \approx 50$ m at 7 Hz, $\gg$ body dimension ~0.3 m).

Two idealized geometries bracket the range of real pocket shapes:

- **Spherical:** radius $a = (3V/4\pi)^{1/3}$, appropriate for isolated gas pockets in distended segments.
- **Cylindrical:** fixed lumen radius $R = 15$ mm (small intestine), length $L = V/(\pi R^2)$, appropriate for elongated gas slugs.

Canonical parameter values are listed in Table 1.

Table 1: Model parameters.

Symbol	Description	Value	Source
γ	Specific-heat ratio (air)	1.4	—
P_0	Ambient (atmospheric) pressure	101 325 Pa	—
ρ_f	Tissue/fluid density	1020 kg/m^3	Fung [1993]
E_w	Intestinal wall Young's modulus	10 kPa	Gregersen and Kassab [2000]
ν_w	Intestinal wall Poisson's ratio	0.45	Fung [1993]
h_w	Intestinal wall thickness	3 mm	Higa et al. [2007]
ρ_w	Wall density	1040 kg/m^3	—
c_f	Longitudinal wave speed in tissue	1540 m/s	—
μ	Dynamic viscosity	1×10^{-3} Pa · s	—
R	Lumen radius (small intestine)	15 mm	—

2.2 Resonance frequency

2.2.1 Spherical pocket: modified Minnaert frequency

The classical Minnaert frequency for a free spherical gas bubble of radius a in an infinite liquid is [Minnaert, 1933]:

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}. \tag{1}$$

For a pocket enclosed by an elastic shell (the intestinal wall), both the gas stiffness and the wall stiffness contribute, while the inertia includes both the radiation added mass of the surrounding fluid and the shell mass [Church, 1995, Hoff et al., 2000]. Expressing all quantities as stiffness and inertia per unit radial displacement:

$$f_0^{(\mathrm{sph})} = \frac{1}{2\pi}\sqrt{\frac{k_{\mathrm{gas}} + k_{\mathrm{wall}}}{m_{\mathrm{fluid}} + m_{\mathrm{wall}}}}, \tag{2}$$

where

$$k_{\mathrm{gas}} = \frac{3\gamma P_0}{a}, \qquad k_{\mathrm{wall}} = \frac{2\,E_w\,h_w}{a^2\,(1-\nu_w)}, \tag{3}$$

$$m_{\mathrm{fluid}} = \rho_f\,a, \qquad m_{\mathrm{wall}} = \rho_w\,h_w. \tag{4}$$

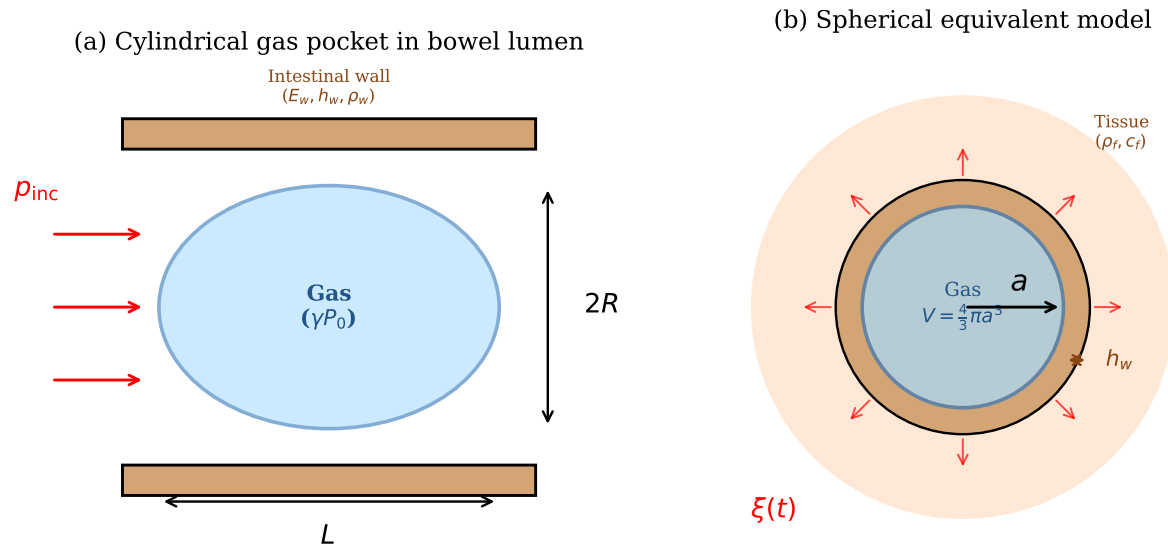

Figure 1: (a) Cylindrical gas pocket constrained within the intestinal lumen. The incident pressure p_{inc} is spatially uniform at infrasound wavelengths. (b) Spherical equivalent model showing radial oscillation $\xi(t)$ of the gas–tissue interface, elastic wall of thickness h_w, and surrounding tissue of density ρ_f.

The gas stiffness arises from adiabatic compression of the enclosed gas (Boyle's law linearised about equilibrium radius a); the wall stiffness from hoop stress in a thin spherical shell; and the inertia terms from radiation loading and shell mass per unit area, respectively. In the limit $h_w \to 0$ (free bubble), k_{wall} and m_{wall} vanish and Eq. (2) reduces exactly to the Minnaert result, Eq. (1):

$$f_0 \to \frac{1}{2\pi}\sqrt{\frac{3\gamma P_0/a}{\rho_f\, a}} = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}} = f_M.$$

For the soft intestinal wall ($E_w = 10\,\text{kPa}$), k_{wall} contributes only 0.9–2.4% of the total effective stiffness $k_{\text{gas}} + k_{\text{wall}}$ across the physiological pocket-size range; gas compressibility dominates. The wall mass $m_{\text{wall}} = \rho_w h_w$ adds ~20% to the fluid inertia, which lowers f_0 slightly below f_M. The "constrained" label therefore refers to the geometric confinement of the gas pocket within the tubular lumen, not to a significant elastic restraint by the wall.

2.2.2 Cylindrical pocket: radial and axial modes

For an elongated gas column in an elastic tube of radius R, the lowest *radial* (breathing) mode has frequency:

$$f_0^{(\text{cyl,rad})} = \frac{1}{2\pi}\sqrt{\frac{2\gamma P_0/R + E_w h_w/[R^2(1-\nu_w^2)]}{\rho_f R + \rho_w h_w}}. \tag{5}$$

The factor of 2 (vs. 3 for the sphere) reflects the cylindrical geometry.

The *axial* (piston) mode treats the gas slug as a spring with stiffness $k = \gamma P_0 \pi R^2/L$ and radiation mass $m_{\text{end}} = \frac{8}{3}\rho_f R^3$ at each open end [Leighton, 1994]:

$$f_0^{(\text{cyl,ax})} = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0\, \pi R^2}{2L \cdot \frac{8}{3}\rho_f R^3}}. \tag{6}$$

The lowest resonance for a cylindrical pocket is $f_0 = \min(f_0^{(\text{cyl,rad})}, f_0^{(\text{cyl,ax})})$.

For physiological gas pocket volumes (1 mL to 100 mL), resonance frequencies range from ~30 Hz (large cylindrical pockets, axial mode) to ~440 Hz (small spherical pockets with elastic wall). Because wall stiffness adds little and wall mass lowers f_0 below the free-bubble Minnaert value, the constrained spherical frequencies are roughly 89–96% of f_M for clinically relevant volumes of 5 mL to 100 mL (Table 2). All resonances remain well above 20 Hz—i.e. above the conventional infrasound boundary. This is a critical distinction from earlier analyses that sought resonance *at* infrasound frequencies.

2.3 Damping

The total damping ratio is

$$\zeta = \tfrac{1}{2}(\delta_{\text{rad}} + \delta_{\text{th}} + \delta_{\text{vis}} + \delta_{\text{wall}}), \tag{7}$$

with contributions from:

$$\delta_{\text{rad}} = \frac{\omega a}{c_f} \qquad \text{(radiation into surrounding tissue)}, \tag{8}$$

$$\delta_{\text{th}} \approx 0.04 \qquad \text{(thermal conduction in gas)}, \tag{9}$$

$$\delta_{\text{vis}} = \frac{4\mu}{\rho_f\, \omega\, a^2} \qquad \text{(viscous boundary layer)}, \tag{10}$$

$$\delta_{\text{wall}} \approx 0.3 \qquad \text{(structural loss in intestinal wall [Fung, 1993])}. \tag{11}$$

2.4 Forced response: sub-resonant transduction

The key mechanism is not *resonance* but *compliance*: gas pockets respond to the incident pressure because they are compressible, even far below their resonant frequency. The radial displacement of the gas–tissue interface under a spatially uniform incident pressure p_{inc} is:

$$\xi = \frac{p_{\text{inc}}}{k_{\text{eff}}} \cdot H(\omega), \tag{12}$$

where k_{eff} is the effective radial stiffness (gas plus wall contributions) and $H(\omega)$ is the standard single-degree-of-freedom frequency response:

$$H(\omega) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_0} = \frac{f}{f_0}. \tag{13}$$

In the sub-resonant regime ($f \ll f_0$, so $r \ll 1$), $H \to 1$ and the displacement becomes frequency-independent:

$$\xi_{\text{sub-res}} \approx \frac{p_{\text{inc}}}{k_{\text{eff}}}. \tag{14}$$

For a spherical pocket with elastic wall:

$$k_{\text{eff}}^{(\text{sph})} = \frac{3\gamma P_0}{a} + \frac{2\,E_w\,h_w}{a^2(1-\nu_w)}. \tag{15}$$

For a cylindrical pocket:

$$k_{\text{eff}}^{(\text{cyl})} = \frac{2\gamma P_0}{R} + \frac{E_w\,h_w}{R^2(1-\nu_w^2)}. \tag{16}$$

The incident pressure at SPL S (in dB re 20 µPa) is:

$$p_{\text{inc}} = p_{\text{ref}} \times 10^{S/20}, \qquad p_{\text{ref}} = 20\,\mu\text{Pa}. \tag{17}$$

2.5 Volume displacement and tissue coupling

The oscillating gas–tissue interface radiates displacement into the surrounding tissue as a monopole source. The volume displacement is:

$$\Delta V = \begin{cases} 4\pi a^2\,\xi & \text{(spherical)}, \\ 2\pi R\,L\,\xi & \text{(cylindrical, radial mode)}. \end{cases} \tag{18}$$

In the near field, the tissue displacement at distance r from the pocket centre decays as:

$$u(r) \approx \left(\frac{a}{r}\right)^2 \xi, \qquad r > a. \tag{19}$$

This concentrates the acoustic displacement at the gas–tissue boundary, precisely where Piezo channels are located in the intestinal wall [Alcaino et al., 2017, Mazzuoli-Weber and Schemann, 2015].

2.6 Illustrative mechanotransduction comparison

To place the predicted displacements in a biological context, we compare them against the activation threshold of Piezo2 mechanosensitive ion channels. Coste et al. [2010] reported membrane indentation thresholds of 0.5 µm to 2.0 µm in patch-clamp experiments on isolated cells; we adopt $\xi_{\text{PIEZO}} = 0.5\,\mu\text{m}$ as an illustrative lower bound.

We emphasise that this comparison is presented to establish *order-of-magnitude plausibility*, not quantitative equivalence. The patch-clamp threshold reflects membrane deformation of individual cells under controlled conditions; the bulk tissue wall displacement computed by the constrained-bubble model involves a fundamentally different strain field. Resolving whether gas-pocket wall displacement can activate Piezo channels *in situ* requires tissue-level mechanotransduction measurements that do not currently exist (see §5).

Subject to this caveat, the minimum SPL for which the model predicts $\xi = \xi_{\text{PIEZO}}$ is:

$$\text{SPL}_{\text{thresh}} = 20\log_{10}\left(\frac{\xi_{\text{PIEZO}}\,k_{\text{eff}}}{p_{\text{ref}}\,H(\omega)}\right). \tag{20}$$

2.7 Sensitivity analysis: population variability

Inter-individual variability in bowel gas content is substantial [Levitt, 1971, Bedell et al., 1956, Serra et al., 2001]. To identify which parameters most affect the predicted displacement, we perform a Monte Carlo sensitivity analysis with $N = 10\,000$ sampled individuals, each characterised by:

- Total bowel gas volume drawn from a log-normal distribution with median 200 mL and $\sigma_{\ln} = 0.65$, giving a 95% CI of approximately 56 mL to 715 mL and a heavy right tail extending to ~1500 mL;
- Number of discrete pockets $N_p \sim \text{Poisson}(\lambda = 10)$, clipped to $[2,\ 40]$;
- Per-pocket volumes drawn from a symmetric Dirichlet partition of the total volume;
- Geometry randomly assigned (70% cylindrical, 30% spherical).

For each sampled individual, the maximum pocket-wall displacement at 7 Hz and 120 dB is computed under the idealised sealed-pocket assumption. This analysis identifies the sensitivity of the outcome to the assumed input distributions; it does not constitute independent validation of the mechanism (see §3.3).

3 Results

3.1 Resonance frequencies are above infrasound

Table 2 summarizes the resonance frequencies for representative gas pocket sizes. Spherical pocket resonances range from approximately 108 Hz at 100 mL to 438 Hz at 1 mL, and the cylindrical radial (breathing) mode lies at ~162 Hz. However, cylindrical *axial* (piston) modes fall below 100 Hz for pockets larger than ~10 mL, reaching 31 Hz at 100 mL—near, but still above, the conventional infrasound boundary (20 Hz). None of the pockets resonate *at* infrasound frequencies; the sub-resonant compliance mechanism described below operates efficiently across the entire infrasound band.

3.2 SPL thresholds under idealised conditions

Figure 2 shows the SPL predicted by the model to produce 0.5 µm wall displacement as a function of gas pocket volume, *assuming acoustically sealed pockets* (i.e. no pressure equalisation via the luminal air column; see §4.2 for the effect of relaxing this assumption). Under these idealised conditions:

- For **spherical** pockets, the predicted threshold decreases from ~120 dB at 5 mL to ~111 dB at 100 mL.
- For **cylindrical** pockets, the predicted threshold is nearly independent of volume (~114 dB), because the lumen radius R is fixed.
- The 9-dB range across physiological pocket sizes maps directly to the range of inter-individual gas content.

Table 2: Resonance frequencies for elastic-wall gas pockets.

V [mL]	Geometry	a or R [mm]	f_{Minnaert} [Hz]	f_0 [Hz]
1	Spherical	6.2	524	438
5	Spherical	10.6	306	273
10	Spherical	13.4	243	222
20	Spherical	16.8	193	179
50	Spherical	22.9	142	134
100	Spherical	28.8	113	108
10	Cylindrical†	15.0	—	99
50	Cylindrical†	15.0	—	44
100	Cylindrical†	15.0	—	31

†Cylindrical f_0 is the axial (piston) mode, which is lowest. The radial (breathing) mode is ∼162 Hz, independent of volume.

These values represent an *upper bound* on the mechanism's efficiency. As discussed in §4.2, acoustic short-circuiting can substantially reduce the effective driving pressure, raising the true thresholds above these sealed-pocket estimates.

3.3 Sensitivity analysis results

Figure 3 presents the Monte Carlo sensitivity analysis for $N = 10\,000$ sampled individuals at 120 dB, under the sealed-pocket assumption. Under these idealised conditions, the model predicts that nearly all sampled individuals would exceed the illustrative 0.5 µm displacement threshold. This high exceedance fraction is a direct consequence of the input assumptions rather than an independent prediction: the model assigns every individual at least 30 mL of bowel gas (a physiologically conservative lower bound), and even a 3 mL cylindrical pocket produces $\xi > 0.5$ µm at 120 dB in the sealed-pocket limit.

The value of this analysis lies not in the exceedance fraction itself—which is tautological given the assumed distributions—but in identifying the *sensitivity* of the outcome to individual parameters. Gas pocket volume emerges as the dominant factor: the distribution of maximum pocket-wall displacement spans approximately 1.03 µm to 2.70 µm, with a median of ∼1.08 µm, tracking the log-normal gas volume distribution closely. Individuals in the upper tail (>500 mL total gas) have predicted displacements exceeding 2 µm. Crucially, however, these values assume perfect acoustic sealing. If the short-circuit attenuation factor of ∼0.17 (see §4.2) is applied, the median displacement falls to ∼0.18 µm—well below the illustrative Piezo threshold. The mechanism therefore operates only if gas pockets are acoustically sealed, making pocket sealing the single most important parameter in the model.

3.4 Comparison with whole-cavity resonance

Figure 4 compares the gas-pocket transduction pathway with the whole-cavity flexural mode ($n = 2$) pathway from Mace et al. [2026], both evaluated under their respective idealised assumptions. Under the sealed-pocket assumption, the gas-pocket mechanism predicts tissue displacement that

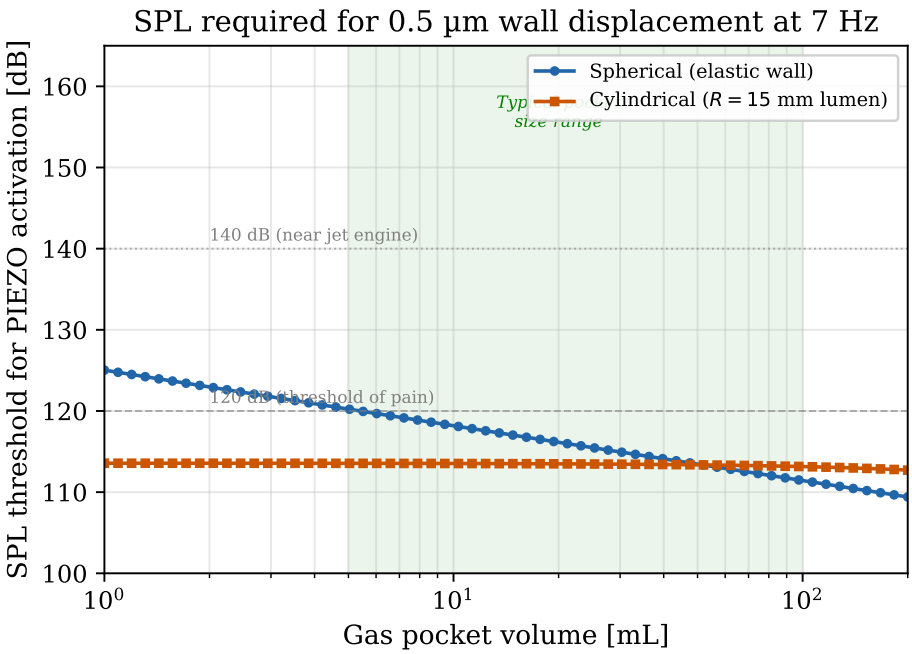

Figure 2: Predicted SPL for 0.5 µm wall displacement (illustrative Piezo threshold) as a function of gas pocket volume at 7 Hz, assuming acoustically sealed pockets (upper-bound estimate; see text). The shaded region indicates the typical physiological pocket size range. Horizontal lines mark 120 dB (threshold of pain) and 140 dB (near jet engine).

is **35–100× larger** than the whole-cavity pathway at the same SPL, across the entire range from 80 dB to 150 dB.

At 120 dB, the whole-cavity $n = 2$ mode yields $\xi \approx 0.014\,\mu\text{m}$ (energy-consistent; Mace et al., 2026)—more than an order of magnitude below the illustrative Piezo threshold. In contrast, a 100 mL sealed gas pocket yields $\xi \approx 1.3\,\mu\text{m}$. If acoustic short-circuit attenuation is applied (factor ~0.17 at 7 Hz), this drops to ~0.22 µm—below the illustrative threshold, but still an order of magnitude above the whole-cavity pathway. The gas-pocket mechanism thus remains the more efficient of the two pathways regardless of the sealing assumption, but whether either pathway reaches biologically relevant amplitudes is an open question.

4 Discussion

4.1 Gas pockets as candidate impedance-matching transducers

The central hypothesis of this paper is that intraluminal gas pockets could serve as *impedance-matching acoustic transducers* between the airborne pressure field and the intestinal wall. The proposed mechanism is not resonance enhancement—the gas-pocket resonances still lie well above the 7 Hz driving frequency, at roughly 31 Hz to 438 Hz across the cases studied—but rather the *compliance mismatch* between the compressible gas and the surrounding incompressible tissue.

At infrasound wavelengths ($\lambda \gg$ body size), the incident pressure is spatially uniform throughout the abdomen. In incompressible tissue, this uniform pressure produces negligible strain (bulk

modulus ~2.2 GPa). But at a gas–tissue interface, the gas compresses while the tissue does not, creating a displacement discontinuity that scales as $\xi \propto p_{\text{inc}}/k_{\text{eff}}$. This is fundamentally the same mechanism by which the middle ear's ossicular chain provides impedance matching between air and the cochlear fluid [Junger and Feit, 1986], albeit at much lower frequencies and without mechanical advantage.

4.2 Acoustic short-circuit: the critical limitation

The most important finding of this analysis is the identification of the *acoustic short-circuit* as the critical factor that determines whether the proposed mechanism can operate.

The GI tract is an open tube (mouth to anus, total length ~8 m). At infrasound wavelengths ($\lambda = c_{\text{air}}/f \approx 49$ m at 7 Hz), $\lambda \gg L_{\text{GI}}$, so luminal air could in principle deliver the same incident pressure p_{inc} to the *interior* of a gas pocket as tissue delivers to its *exterior*. If both paths transmit p_{inc} faithfully, the net pressure difference across the gas–tissue interface vanishes: $\Delta P \longrightarrow 0$, and no displacement results—an *acoustic short circuit*. The gut, it seems, has evolved an elegant defence against its own acoustics.

We estimate the cross-over frequency by modelling the open GI tract as a Helmholtz resonator. The luminal gas volume V communicates with the exterior through a constricted air column of cross-section S and effective acoustic length L_{eff}. The Helmholtz resonance frequency is

$$f_H = \frac{c_{\text{air}}}{2\pi}\sqrt{\frac{S}{V\,L_{\text{eff}}}}. \tag{21}$$

For representative values ($V \approx 200$ mL, $S \approx \pi(5\,\text{mm})^2$, $L_{\text{eff}} \approx 5$ m), this gives $f_H \approx 15$ Hz. Below f_H, the air column can quasi-statically equalise pressure, partially short-circuiting the transduction mechanism; above f_H, air-column inertia prevents equalisation and the constrained-bubble model applies without qualification.

At 7 Hz ($r = f/f_H \approx 0.47$), the equalisation parameter $E = 1/[1 + (f/f_H)^2] \approx 0.83$, indicating that ~83% of the exterior pressure is transmitted to the pocket interior, leaving only ~17% of p_{inc} as the effective driving pressure. This attenuation is devastating for the mechanism: the strongest pocket considered (100 mL, spherical, $\xi = 1.35$ µm under sealed conditions) drops to $\xi \approx 0.23$ µm—below the illustrative Piezo threshold of 0.5 µm.

We note that Eq. (21) is used here only to estimate f_H as a cross-over scale; the implied low-pass attenuation model $1/[1 + (f/f_H)^2]$ is a simplification of the full Helmholtz transfer function. A true Helmholtz resonator exhibits amplification near f_H and a steeper roll-off above it; moreover, the effective path length and aperture diameter governing f_H are poorly constrained anatomically. The true equalisation may therefore be *worse* (~90%) than the 83% estimated here, which would strengthen rather than weaken the short-circuit conclusion. This warrants future investigation with anatomically informed impedance models.

4.2.1 Bounded analysis: sealed vs. open pockets

The results presented in §3 assume acoustically sealed pockets (no interior pressure equalisation). This represents an *upper bound* on the mechanism's efficiency. The short-circuit analysis above provides an approximate *lower bound* for open pockets. The true situation lies between these extremes, and depends critically on the degree of acoustic sealing *in vivo*.

We acknowledge a tension in the boundary conditions employed in this paper. The cylindrical axial (piston) resonance mode (Eq. 6) assumes open tube ends (pressure-release boundaries), which is appropriate for computing resonance frequencies of a gas slug communicating with adjacent fluid. However, the defence against the acoustic short-circuit relies on pockets being sealed by liquid or fecal plugs. These are mutually contradictory assumptions applied in different parts of the analysis to obtain favourable results in each case. We present both cases transparently as bounds rather than claiming one is correct:

- **Sealed pocket (upper bound):** The luminal air path is blocked by chyme, mucus, or fecal plugs. The constrained-bubble model applies in full. Imaging studies confirm that gas is typically distributed in discrete, separated pockets rather than a single connected volume [Levitt, 1971, Serra et al., 2001], and healthy fasting individuals harbour 5–15 discrete pockets separated by liquid or collapsed lumen [Suarez et al., 1997]. Under this assumption, the maximum displacement for a 100 mL pocket at 120 dB is $\xi \approx 1.35\,\mu\text{m}$.

- **Open pocket (lower bound):** The pocket communicates freely with the GI air column. The short-circuit transfer function attenuates the effective driving pressure by ~83% at 7 Hz, reducing the displacement to $\xi \approx 0.23\,\mu\text{m}$, below the illustrative Piezo threshold.

The acoustic sealing of gas pockets *in vivo* is therefore the single most important parameter in determining whether this mechanism operates. No published measurements of *in vivo* pocket impedance or acoustic transmission through intestinal liquid plugs exist. Obtaining such measurements is the highest-priority experimental requirement identified by this analysis (see §5).

4.3 Clinical implications (conditional)

If the gas-pocket mechanism operates *in vivo* (i.e. if pockets are sufficiently sealed), the model predicts that individuals with elevated bowel gas would be more susceptible to infrasound-induced GI effects. Several clinical populations have chronically elevated intestinal gas:

- **Irritable bowel syndrome (IBS):** Patients with IBS exhibit altered gas handling and increased gas retention [Serra et al., 2001], as well as visceral hypersensitivity that may lower the effective Piezo threshold.

- **Post-prandial states:** Gas production increases significantly after meals, particularly with fermentable substrates, transiently increasing both total gas volume and the number of discrete pockets [Harder et al., 2003].

- **Aerophagia:** Excessive air swallowing introduces large gas volumes into the upper GI tract.

- **Small intestinal bacterial overgrowth (SIBO):** Excess fermentation produces elevated luminal gas.

This framework would naturally explain the sporadic and individually variable nature of reported infrasound effects: susceptibility would depend on a physiological parameter (gas content and pocket sealing) that varies by up to 15× across the population and fluctuates within individuals on hourly timescales. We stress, however, that these clinical implications remain contingent on the mechanism operating at all, which requires experimental resolution of the acoustic sealing question.

4.4 Comparison with established bubble dynamics

Our constrained-bubble model builds on the extensive literature on encapsulated microbubbles used as ultrasound contrast agents [Church, 1995, Hoff et al., 2000]. The key differences are:

1. **Scale:** Ultrasound contrast agents are ~1 µm to 10 µm in diameter; bowel gas pockets are 1 mm to 60 mm.

2. **Wall properties:** Contrast agent shells are polymeric ($E \sim 1$ GPa); the intestinal wall is a soft biological tissue ($E \sim 10$ kPa).

3. **Frequency regime:** Contrast agents are driven near resonance (1 MHz to 10 MHz); bowel gas pockets are driven far below resonance ($f/f_0 \sim 0.01$–0.05).

Despite these differences, the underlying physics is identical: pressure-driven radial oscillation of a compressible inclusion in an elastic medium. The sub-resonant regime simply replaces the resonance-amplified response with a quasi-static compliance-dominated response.

4.5 Limitations and key unknowns

Beyond the acoustic short-circuit discussed in §4.2, several additional simplifications warrant discussion:

1. **Linear model:** We assume small-amplitude oscillations. At very high SPL (> 140 dB), nonlinear effects (rectified diffusion, microstreaming) may become relevant.

2. **Idealised geometries:** Real gas pockets have irregular shapes. Our spherical and cylindrical models bracket the physiological range but do not capture complex topologies. The cylindrical breathing-mode model assumes $L/R \geq 1$; for pockets smaller than ~10 mL in a 15 mm lumen, the spherical model is more appropriate as the pocket geometry becomes disc-like.

3. **Quasi-static tissue response:** The monopole coupling model (Eq. 19) neglects viscoelastic relaxation, which would reduce the displacement at longer distances.

4. **Threshold category error:** As discussed in §2.6, the Piezo activation threshold of 0.5 µm derives from *in vitro* patch-clamp experiments on isolated cells and measures membrane indentation, not bulk tissue wall displacement. These are fundamentally different strain fields, and the comparison is presented as illustrative only. The *in vivo* threshold for Piezo activation by gas-pocket wall oscillation is unknown.

5. **No experimental validation:** The model predictions remain entirely untested. The theoretical framework identifies what measurements are needed (§5), but until those measurements are performed, the mechanism remains a hypothesis.

6. **Boundary condition inconsistency:** As noted in §4.2, the axial resonance mode assumes open ends while the short-circuit defence assumes sealed pockets. We present both cases as bounds rather than resolving this tension, which requires experimental data.

5 Experimental requirements

The theoretical framework developed above identifies three critical measurements that would resolve whether the gas-pocket mechanism operates *in vivo*. We present these in order of priority.

5.1 In-vivo pocket impedance and acoustic sealing

The analysis identifies pocket acoustic sealing as the single most important determinant of whether the mechanism can operate. The required measurement is the acoustic transmission coefficient through the liquid or fecal plugs that separate gas pockets from the continuous GI air column.

Proposed approach: Simultaneous manometry (intraluminal pressure) at two locations flanking a liquid plug, with controlled acoustic excitation at infrasound frequencies (1 Hz to 20 Hz). The ratio of transmitted to incident pressure amplitude as a function of frequency would directly determine the equalisation parameter E and hence the effective short-circuit attenuation.

Minimum requirement: If the measured attenuation through a typical liquid plug exceeds 15 dB at 7 Hz (i.e. $< 18\%$ pressure transmission), the sealed-pocket assumption is justified and the constrained-bubble model applies in full. If the attenuation is less than 5 dB, the short circuit is dominant and the mechanism is effectively suppressed.

5.2 Tissue-level mechanotransduction thresholds

The 0.5 µm threshold used in this paper derives from single-cell patch-clamp experiments and is not directly comparable to bulk tissue wall displacement. Resolving this requires *in vitro* experiments on intestinal tissue explants (not isolated cells) subjected to controlled oscillatory strain at infrasound frequencies.

Proposed approach: Excised intestinal wall segments mounted in a perfusion chamber with piezoelectric actuators delivering controlled sinusoidal displacement at 1 Hz to 20 Hz. Readout via calcium imaging or electrophysiology of the enteric nervous system response. The threshold displacement for detectable neural activation would provide the tissue-level equivalent of the single-cell Piezo threshold.

5.3 Pocket geometry under acoustic excitation

Real-time imaging of gas pocket geometry during infrasound exposure would validate the assumed spherical/cylindrical models and measure actual wall displacements.

Proposed approach: Dynamic MRI or high-frame-rate ultrasound imaging of abdominal gas pockets during controlled infrasound exposure at 100 dB to 120 dB. The imaging would need temporal resolution ≤ 50 ms (at least 3 frames per cycle at 7 Hz) and spatial resolution ≤ 0.5 mm to detect the predicted sub-micrometre wall displacements—a significant technical challenge that may require phase-sensitive ultrasound techniques.

6 Conclusion

This work establishes a theoretical framework for evaluating the hypothesis that intraluminal gas pockets in the gastrointestinal tract could act as acoustic transducers, converting airborne

infrasonic pressure to local wall strain via a sub-resonant compliance mechanism.

Under idealised sealed-pocket conditions, the constrained-bubble model predicts tissue displacements 35–100× larger than the whole-cavity flexural resonance pathway, with SPL thresholds for reaching an illustrative mechanotransduction level ranging from 111 dB to 120 dB. However, the analysis identifies two critical limitations that prevent these predictions from being presented as established findings:

1. **Acoustic short-circuiting** via the GI air column attenuates the effective driving pressure by up to 83% at 7 Hz for open pockets, reducing all predicted displacements below the illustrative threshold. The mechanism operates only if gas pockets are acoustically sealed by mucosal folds, liquid plugs, or luminal collapse—a condition that is physiologically plausible but experimentally uncharacterised.

2. **The mechanotransduction comparison** between bulk tissue wall displacement and single-cell patch-clamp thresholds involves fundamentally different strain fields and is presented as illustrative only.

The analysis identifies pocket acoustic sealing as the critical determinant and defines three experimental measurements—*in vivo* pocket impedance, tissue-level mechanotransduction thresholds, and pocket geometry imaging under acoustic excitation—that would resolve whether this mechanism operates. Pending these measurements, the gas-pocket transduction hypothesis remains the most efficient of the candidate mechanisms analysed here for airborne-infrasound-induced GI effects, but it has not been validated.

If the mechanism is confirmed experimentally, the model makes several testable predictions:

1. SPL thresholds for GI effects should correlate with bowel gas content (measurable by abdominal X-ray or CT);

2. Post-prandial subjects should be more susceptible than fasting subjects;

3. Simethicone (which reduces gas bubble size) should attenuate effects;

4. The effect should be broadband (not frequency-specific), with magnitude scaling as $p_{\mathrm{inc}}/k_{\mathrm{eff}}$. The brown note, if it exists at all, is not a note—it is a chord.

Acknowledgements

Computational assistance was provided by GitHub Copilot CLI (Claude Opus 4), which contributed to code development, literature review, and manuscript preparation.

Data availability

All computational code is available at https://github.com/JonathanMace/brownnote.

Declaration of competing interest

The authors declare no competing interests.

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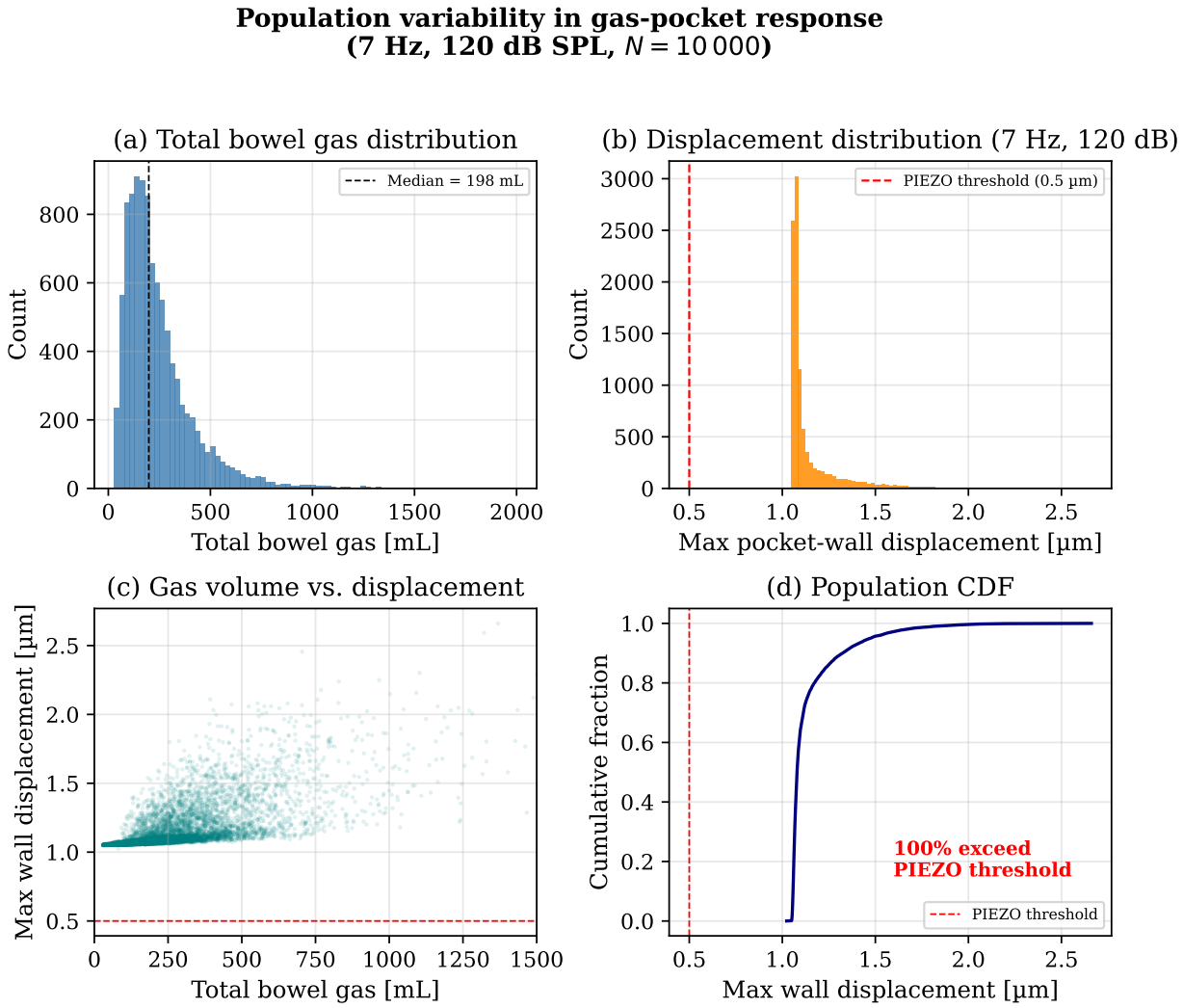

Figure 3: Sensitivity analysis: population variability in gas-pocket acoustic response (7 Hz, 120 dB, $N = 10\,000$, sealed-pocket assumption). (a) Log-normal distribution of total bowel gas. (b) Histogram of maximum pocket-wall displacement. (c) Scatter plot showing the correlation between gas volume and displacement. (d) Cumulative distribution function with the illustrative Piezo threshold marked. These values represent the sealed-pocket upper bound; acoustic short-circuit attenuation would reduce all displacements (see text).

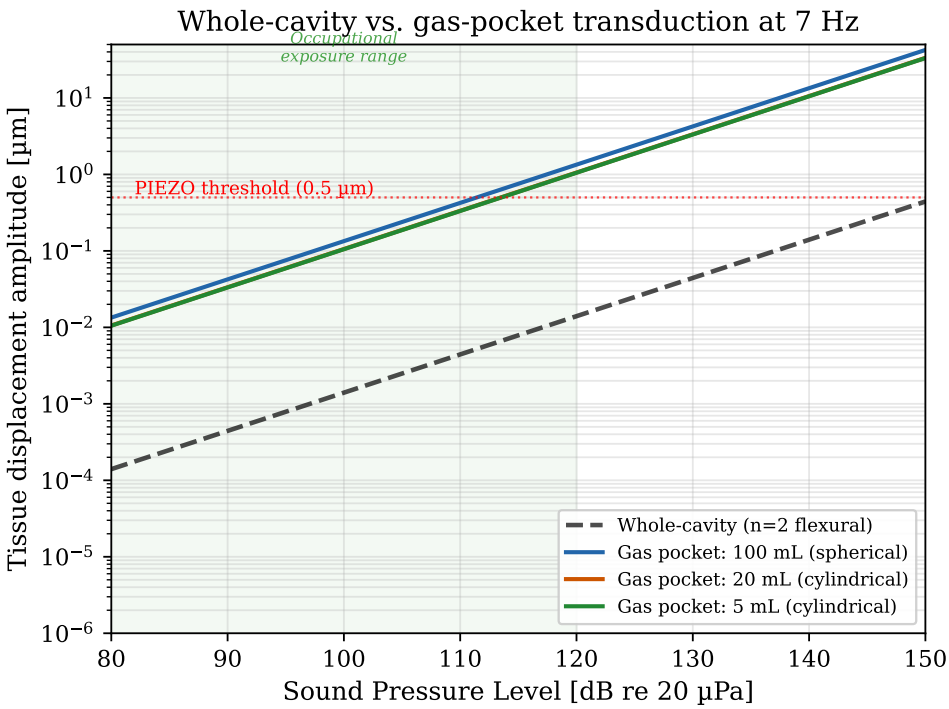

Figure 4: Predicted tissue displacement vs. SPL at 7 Hz: gas-pocket transduction under sealed-pocket assumption (solid lines) vs. whole-cavity flexural resonance (dashed). Under idealised conditions, gas pockets reach the illustrative Piezo threshold (dotted red line) at ~111 dB to 120 dB, while the whole-cavity path requires > 180 dB. Acoustic short-circuit attenuation would shift the solid lines downward (see text).